File permissions in Linux

Project description

The research team at my organization needs to update the file permissions for files and directories within the projects directory. The permissions do not reflect the level of authorization that should be given. Checking and updating these permissions will help keep their system secure. I completed this task by doing:

Check file and directory details

The following code shows how I used Linux commands to determine the existing permissions set for a directory in the file system.

```
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_t.txt
researcher2@c456e5941586:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:51 ..
-rw--w---- 1 researcher2 research_team   46 Jan  8 17:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  8 17:27 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jan  8 17:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jan  8 17:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_t.txt
researcher2@c456e5941586:~/projects$
```

This screenshot displays the command I entered, and the other lines display the output of the command. The code lists all the contents of the projects directory. I used the ls command with the -la option to display a listing of the file contents that also restored hidden files as well. This shows that there is a directory named 'drafts', one hidden file named '.project_x.txt', and five other project files. The 10-character string in the first column shows the permissions set on each file or directory.

Describe the permissions string.

The 10-character string shows who has authorization to access what. The characters each represent something that tells us who has access to what. The first character is either a d or a -, and this indicates the file type. If it's a d it's a directory, and if it's a -, then it's a regular file. The 2nd-4th characters indicate the permission of the user. If the user has full access, it will read rwx. However, if there is a - in place of one of those letters, that means that the user does not have access to that certain permission. R stands for read, w stands for write, and x stands for execute. Characters 5-7 follow this trend, but they represent the permissions for the group. Characters 8-10 also follow this trend, but they represent the permission for other.

Change file permissions

```
researcher2@c456e5941586:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jan  8 17:27 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jan  8 17:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jan  8 17:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_t.txt
researcher2@c456e5941586:~/projects$ chmod o-w project_k.txt
researcher2@c456e5941586:~/projects$ ls
drafts  project_k.txt  project_m.txt  project_r.txt  project_t.txt
researcher2@c456e5941586:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jan  8 17:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jan  8 17:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_t.txt
researcher2@c456e5941586:~/projects$
```

The lines of the screenshot that show the commands chmod o-w project_k.txt and ls -l are the commands that I entered. The chmod command changes the permissions on files and directories. The first argument indicates what permissions should be changed, and the second argument specifies the file or directory. In this screenshot, I removed write permissions from other for the project_k.txt file. After this, I used ls -l to review the updates I made.

Change file permissions on a hidden file

```
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:51 ..
-rw--w---- 1 researcher2 research_team   46 Jan  8 17:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  8 17:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_k.txt
-rw------- 1 researcher2 research_team   46 Jan  8 17:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_t.txt
researcher2@c456e5941586:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@c456e5941586:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:51 ..
-r--r----- 1 researcher2 research_team   46 Jan  8 17:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  8 17:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_k.txt
-rw------- 1 researcher2 research_team   46 Jan  8 17:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_t.txt
researcher2@c456e5941586:~/projects$
```

In this screenshot,t it displays the commands I entered to change the permissions on the hidden file of .project_x.txt. It is a hidden file because it starts with a period. I removed write permissions from the user and group and added read permissions to the group. I removed write permissions from the user with u-w. Then I removed write permissions from the group with g-w, and added read permissions to the group with g+r.

Change directory permissions

```
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:51 ..
-r--r----- 1 researcher2 research_team   46 Jan  8 17:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  8 17:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_k.txt
-rw------- 1 researcher2 research_team   46 Jan  8 17:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_t.txt
researcher2@c456e5941586:~/projects$ chmod g-x drafts
researcher2@c456e5941586:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  8 17:51 ..
-r--r----- 1 researcher2 research_team   46 Jan  8 17:27 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Jan  8 17:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_k.txt
-rw------- 1 researcher2 research_team   46 Jan  8 17:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jan  8 17:27 project_t.txt
researcher2@c456e5941586:~/projects$
```

The screenshot here displays the permission listing for multiple files and directories. I used the code chmod g-x drafts to remove the group's permission to execute in the directory titled drafts. Now in the directory drafts, only the user who is researcher2 can read, write, and execute. Neither the group or other has the permission to do anything.

Summary

I changed multiple permissions to match the level of authorization that my organization wanted for files and directories in the projects directory. The first step in this was using ls -la to check the permissions for the directory. This informed my decisions in the following steps. I then used the chmod command multiple times to change the permissions on files and directories.